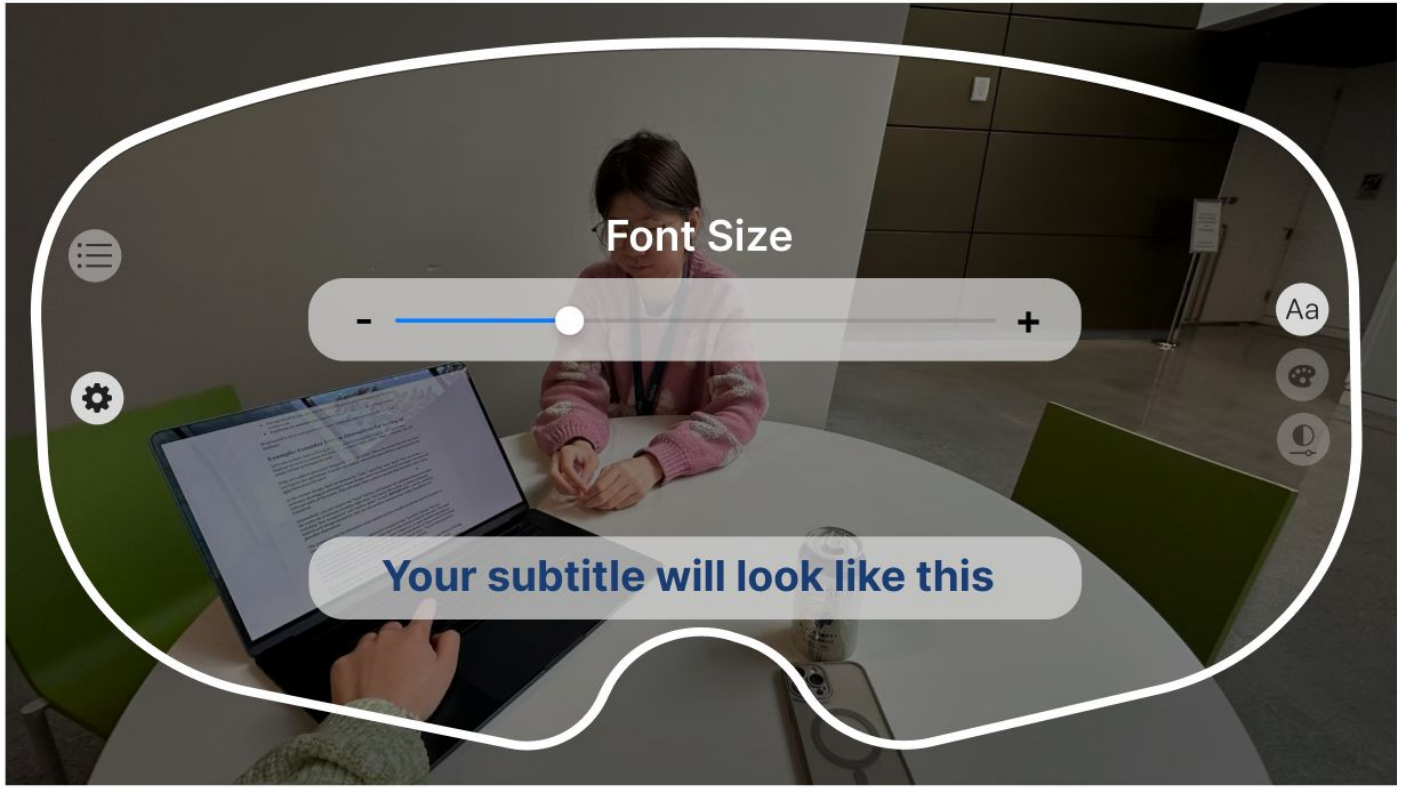
HW5: Updated Vertical Prototype

SignBridge Duo Glass

INFO 5265: Accessibility and Inclusive Design

Group 3: Yuhan Zhang (yz3434), Jiawen Chen (jc3785), Mingyuan Pang (mp2335), August Wang (zw843)

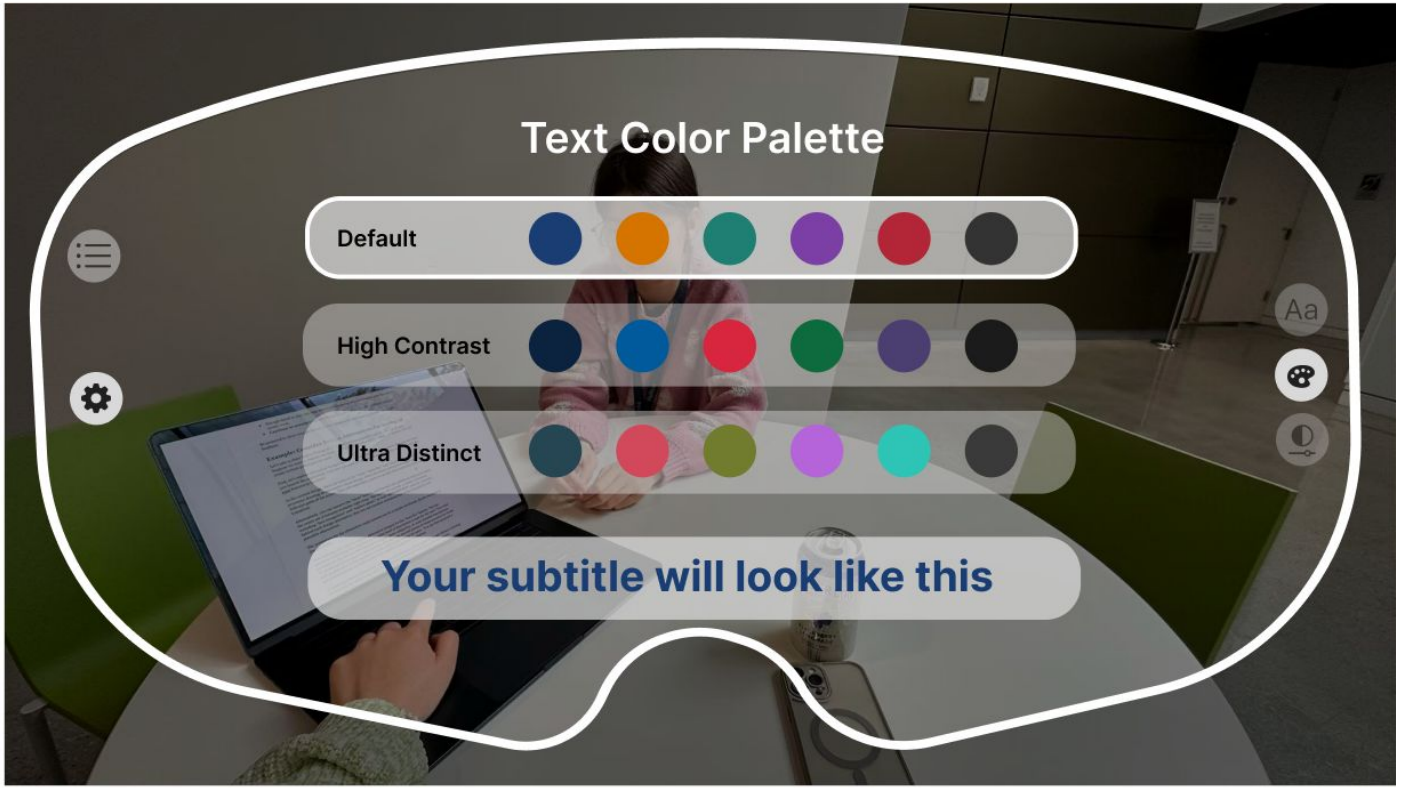
New Feature: Scene 0 - Customization: Font Size



Jiawen, the wearer, activates the on-frame font size adjustment feature while sitting across from her classmate. Using the physical buttons located on the side of the glasses' frame, she scrolls through the size slider that appears in her field of view. As she taps the buttons, the slider smoothly updates, allowing her to preview how subtitles will look in real time.

Feature 0: Customization - Font Size

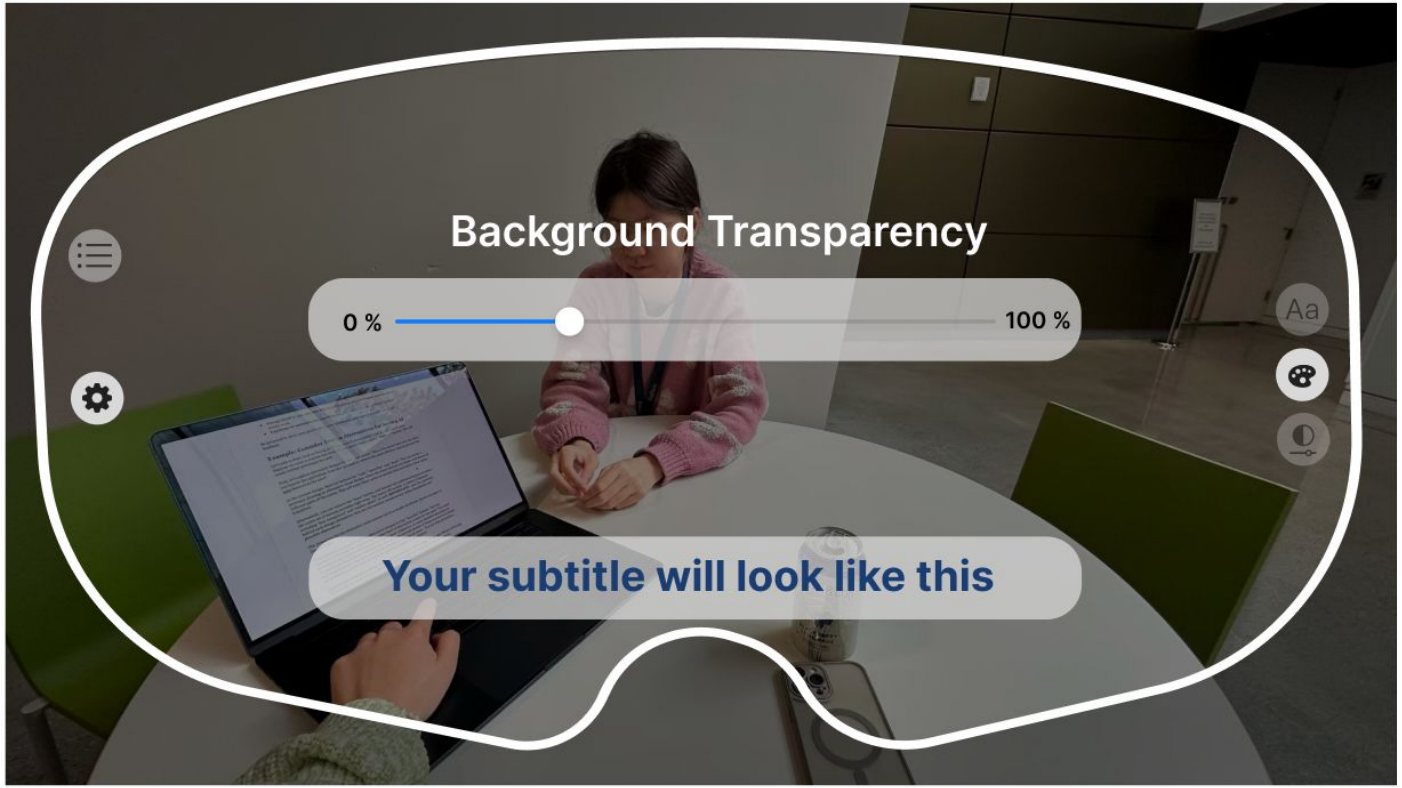
New Feature: Scene 0 - Customization: Text Color Palette



Jiawen then switches to the change Text Color Palette feature. The system displays three preset palettes—Default, High Contrast, and Ultra Distinct—each showing clearly separated color options designed for different accessibility needs. As she previews the palettes, the sample subtitle below updates in real time, letting her see exactly how captions will appear in her view.

Feature 0: Customization - Text Color Palette

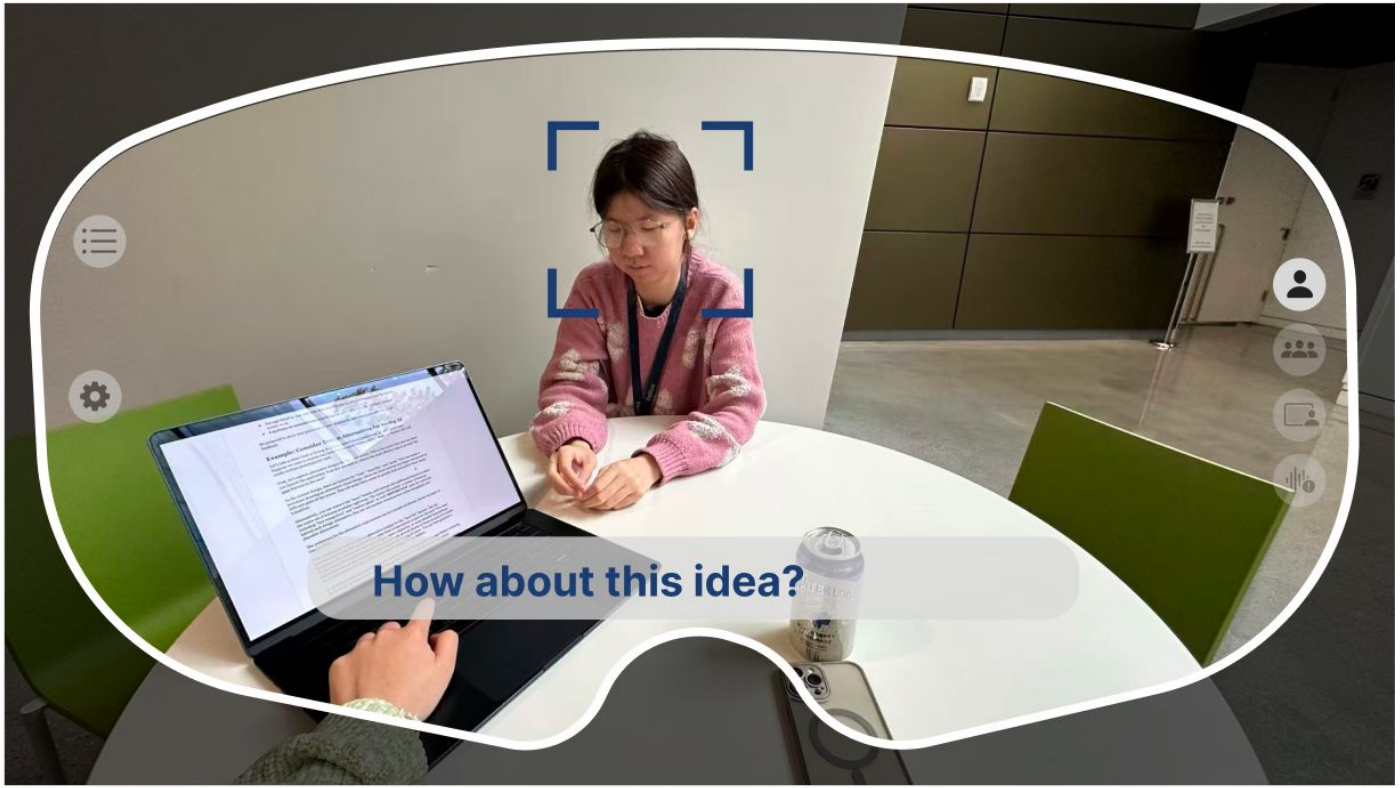
New Feature: Scene 0 - Customization: Background Transparency



Jiawen opens the Background Transparency setting to adjust how opaque the caption bar appears in her view. Using the side buttons on the glasses frame, she slides the transparency level from 0% to 100%, watching the caption background become lighter or more visible in real time. The live subtitle preview updates instantly, helping her find a balance that remains readable while keeping the environment in clear view.

Feature 0: Customization - Background Transparency

Scene 1 — One-on-One Conversation



Jiawen, the wearer, faces Iris in the Bloomberg Level 0 study area. The system automatically detects Iris, highlights her with a blue frame labeled “Iris–friend,” and displays her speech, *“How about this idea?”*, in a centered translucent caption bar. The system started with 1-on-1 mode automatically.

Feature 1: Live Captioning

Scene 2 — Group Greeting

Jiawen waves to Yuhan and August as they join the table. The glasses detect multiple new speakers, each outlined with a distinct colored frame, purple for Yuhan. As Yuhan greets, the system displays his speech, *"Hi, how are you?"*, in a matching purple caption bar centered near the bottom.

Feature 1: Live Captioning

Scene 3 — Group Discussion

Jiawen participates in a group conversation with Iris, Yuhan, and August. Two speakers talk simultaneously, Yuhan (purple frame) says *"I think this is great!"* while August (yellow frame) responds *"I agree!"*. To manage overlapping dialogue, the glasses display an enlarged multi-line caption box at the bottom, color-coded to match each speaker. No manual switching is needed; the system automatically transitions from 1-on-1 mode to group mode.

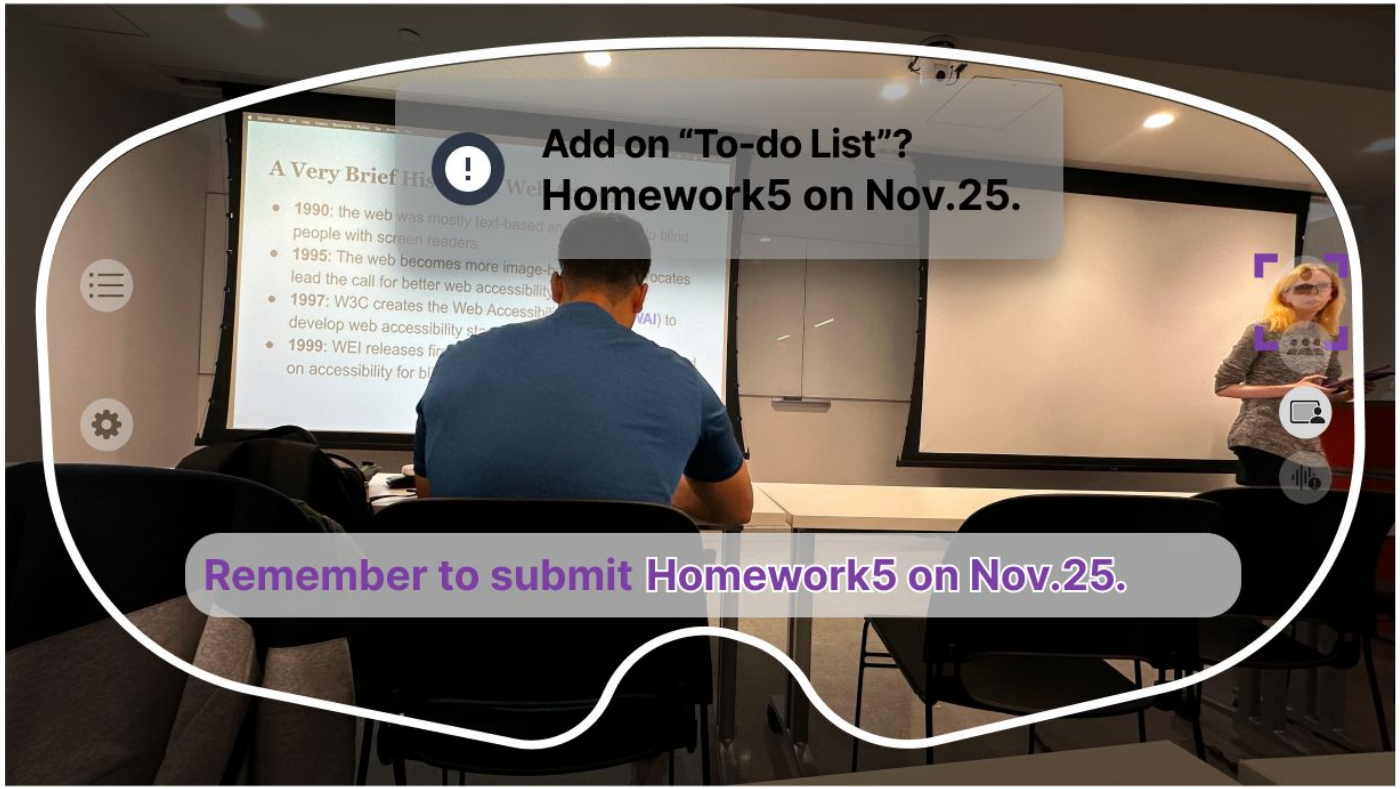
Feature 1: Live Captioning

Scene 4 — Lecture Presentation

Jiawen attends the Accessibility lecture. The professor, standing at the front right, is automatically detected and outlined with a red frame. The glasses localize her voice and display the caption *“Today, we are going to talk about”* in a red bar aligned to the speaker’s direction. No manual control is needed; the system automatically transitions from group mode to presentation mode.

Feature 1: Live Captioning

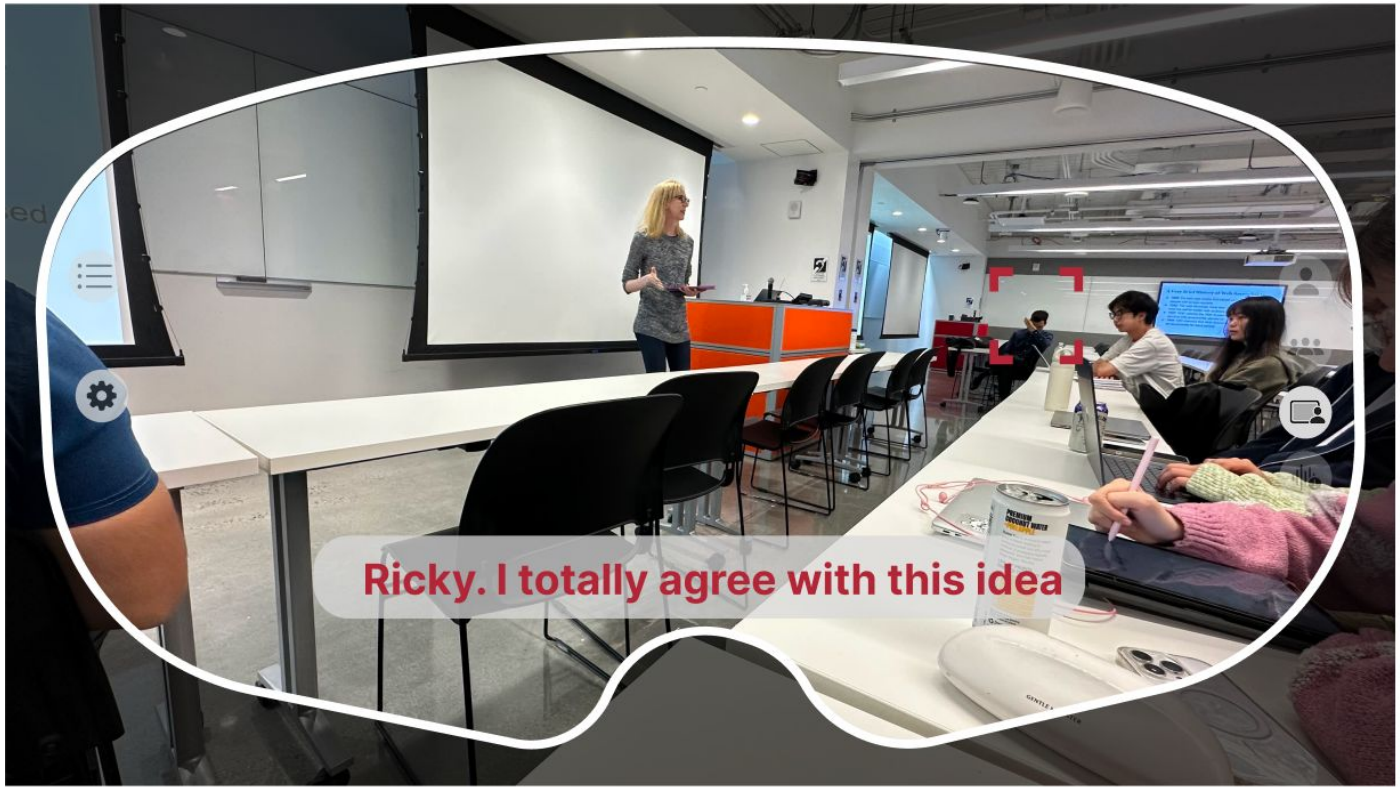
New Feature: Scene 4 — Lecture Presentation: Add to Calendar



During the lecture, the professor’s voice triggers the system’s important-information classifier.
As soon as she says, “Remember to submit Homework 5 on Nov. 25,” the glasses highlight the caption in a purple emphasis bar across the lower field of view.
A soft pop-up card appears in the user’s upper-center vision, asking, “Add on ‘To-do List’? Homework 5 on Nov. 25,” allowing double-click confirmation without disrupting attention. The card will disappear in 10 seconds.

Feature 2: Auto-Important Information Detection

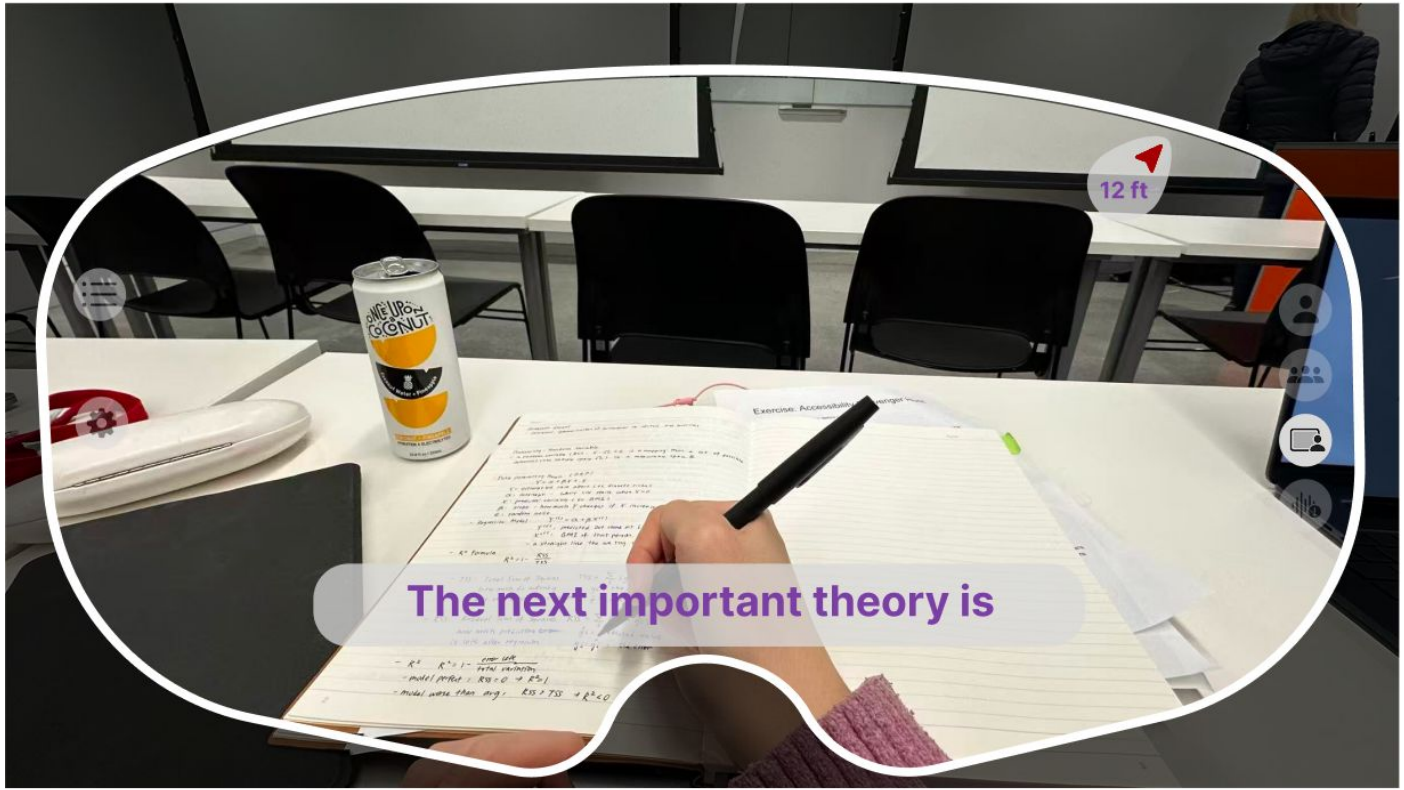
Scene 5 — Class Participation



During the lecture, a distant person participates from the far right side of the classroom. The glasses detect the voice source from a distance and use orange to label his face to indicate his location visually. His speech, *"Ricky, I totally agree with this idea"*, appears in an orange caption bar aligned toward his location.

Feature 1: Live Captioning

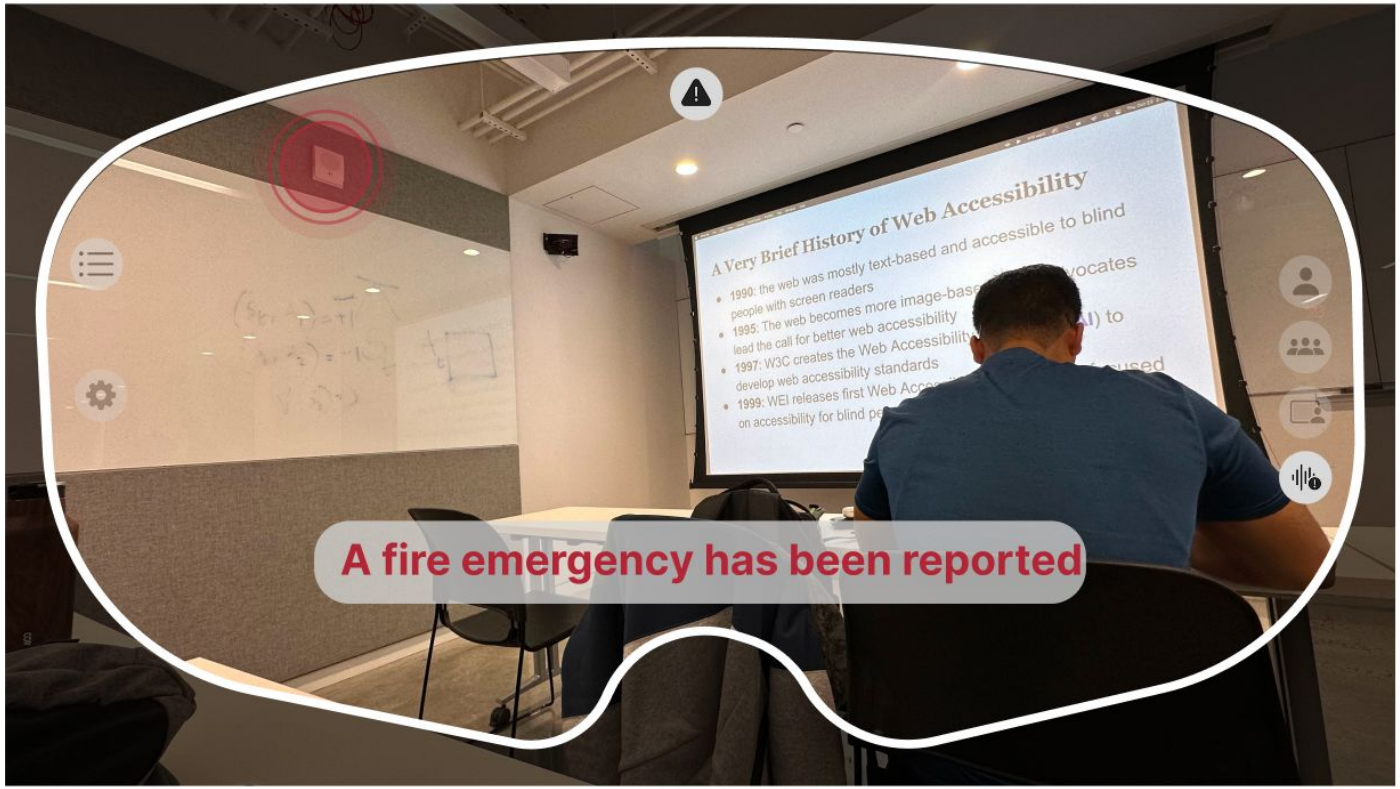
Scene 6 — Note-Taking During Lecture



While taking notes, Jiawen looks down at her notebook as the lecture continues in the background.
The glasses detect that the instructor is speaking from a distance and overlay a subtle red distance marker — “12 ft” — in the upper-right of her field of view, indicating both direction and approximate range to the active speaker.
At the same time, the caption “The next important theory is” appears in a soft purple bar across the lower portion of the lens, ensuring she can follow the explanation without looking up from her handwriting.

Feature 3(Updated): Distance Implicator

Scene 7 — Fire Alarm Alert



While the lecture continues, the ceiling fire alarm activates. The glasses immediately detect the flashing red light and trigger an emergency alert overlay. A blinking red icon appears near the alarm source, and a bold red caption, *"A fire emergency has been reported"*, displays across the center of the lens. No manual input is needed; the system automatically prioritizes safety alerts over other captions and switch to environmental mode.

Feature 4(Updated): Environment Detection

Scene 7 – Fire Alarm Alert

The fire alarm continues to sound as students begin leaving the classroom. The glasses continue to detect an environmental alarm in the building's audio stream. A high-priority alert badge appears at the top of her view, drawing attention without overwhelming the scene.

Feature 4(Updated): Environment Detection